Finite-dimensional counterexamples to a projection-band inverse-limit question

Automated mathematics run `fa_banach_001`
Lane 14, model GPT5.6

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Status: candidate counterexample, likely valid.

1 Source question

Van Amstel and van der Walt introduce, for an Archimedean vector lattice E and a non-trivial ideal M in the Boolean algebra $\mathbb{B}_E$ of projection bands, the inverse system $\mathcal{I}_{\mathrm{M}}$ formed by the bands in M and the band projections between them. Its canonical map is

$$P_{\mathrm{M}} : E \longrightarrow \varprojlim \mathcal{I}_{\mathrm{M}}, \qquad u \longmapsto (P_B u)_{B \in \mathrm{M}}.$$

After Remark 4.12 they ask the following question [1].

Then $\mathrm{M} := \{\mathrm{B}_F \ : \ \varnothing \neq F \subseteq \mathbb{N}$ *is finite*$\}$ *is a proper, non-trivial ideal in* $\mathbf{B}_{\ell^1}$ *and* $\varprojlim \mathcal{I}_{\mathrm{M}}$ *is* $\mathbb{R}^\omega$. *In this case,* $P_{\mathrm{M}}[\ell^1]$ *is a proper subspace of* $\varprojlim \mathcal{I}_{\mathrm{M}}$.

Based on Remark 4.12 we ask the following question: Given a Dedekind complete vector lattice E, does there exist a proper ideal M in $\mathbf{B}_{\mathrm{E}}$ so that $P_{\mathrm{M}} : \mathrm{E} \to \varprojlim \mathcal{I}_{\mathrm{M}}$ is an isomorphism onto $\varprojlim \mathcal{I}_{\mathrm{M}}$? We do not pursue this question any further here, except to note the following example.

> Given a Dedekind complete vector lattice E, does there exist a proper ideal M in $\mathbb{B}_E$ so that $P_{\mathrm{M}} : E \to \varprojlim \mathcal{I}_{\mathrm{M}}$ is an isomorphism onto $\varprojlim \mathcal{I}_{\mathrm{M}}$?

The question occurs on page 23 of the arXiv PDF, immediately after Remark 4.12.

2 Definitions

A *projection band* B of E is a band for which $E = B \oplus B^d$; the associated band projection is denoted by P_B. The projection bands form a Boolean algebra $\mathbb{B}_E$. An ideal $\mathrm{M} \subseteq \mathbb{B}_E$ is downward closed and upward directed. These two properties imply closure under finite joins: if $B, C \in \mathrm{M}$, choose $D \in \mathrm{M}$ with $B, C \subseteq D$ and use downward closure to obtain $B \vee C \in \mathrm{M}$.

For $A \subseteq \{1, \ldots, n\}$, write

$$B_A := \{x \in \mathbb{R}^n : x_j = 0 \text{ for every } j \notin A\}.$$

With the coordinate order, the projection bands of $\mathbb{R}^n$ are exactly the B_A, and P_{B_A} is coordinate restriction to A.

3 Proof intuition

The proposed reconstruction can only be injective if the projections in the ideal separate points. In a finite Boolean algebra, however, a proper ideal has a largest member. For the coordinate bands of $\mathbb{R}^n$, that largest member omits at least one coordinate. The basis vector in the omitted coordinate is invisible to every projection in the ideal. Thus the finite-dimensional case supplies a direct obstruction before surjectivity is even considered.

4 Counterexample

Theorem 1. *For every integer $n \geq 1$, let $E = \mathbb{R}^n$ with its coordinate order. Then E is Dedekind complete, and for every proper ideal $\mathrm{M} \subsetneq \mathbb{B}_E$ the canonical map*

$$P_{\mathrm{M}} : E \longrightarrow \varprojlim \mathcal{I}_{\mathrm{M}}$$

is not injective. Consequently the question as stated has a negative answer.

Proof. Coordinatewise suprema show directly that $E = \mathbb{R}^n$ is Dedekind complete. Identify $\mathbb{B}_E$ with the finite Boolean algebra $\mathcal{P}(\{1, \ldots, n\})$ through $A \mapsto B_A$.

Let M be a proper ideal and put

$$U := \bigcup \{A : B_A \in \mathrm{M}\}.$$

Because $\mathbb{B}_E$ is finite and M is closed under finite joins, $B_U \in \mathrm{M}$. If $U = \{1, \ldots, n\}$, then $B_U = E$ belongs to M; downward closure would give $\mathrm{M} = \mathbb{B}_E$, contradicting properness. Hence some j does not belong to U.

For every $B_A \in \mathrm{M}$ we have $A \subseteq U$, and therefore $P_{B_A} e_j = 0$. By the definition of the canonical map,

$$P_{\mathrm{M}}(e_j) = (P_B e_j)_{B \in \mathrm{M}} = 0.$$

Since $e_j \neq 0$, the map P_{M} is not injective and hence cannot be a lattice isomorphism onto the inverse limit. □

Proposition 1. *More generally, if a nonzero Dedekind complete vector lattice has only finitely many projection bands, no proper ideal of its projection-band Boolean algebra can yield an injective canonical map.*

Proof. The finite ideal has a largest member C, and properness gives $C \neq E$. Choose $0 \neq x \in C^d$. Every band B in the ideal satisfies $B \subseteq C$, so $P_B x = 0$. Hence $P_{\mathrm{M}} x = 0$. □

5 Verification report

The hypotheses and conclusion are checked directly:

1. $\mathbb{R}^n$ is Archimedean and Dedekind complete under coordinatewise order.
2. Its projection bands are the coordinate bands B_A.
3. A finite proper ideal has a largest member $B_U \neq E$.
4. An omitted coordinate vector e_j lies in the kernel of every projection indexed by the ideal, and hence in $\ker P_{\mathrm{M}}$.

No numerical experiment or external theorem is used. If the word “ideal” in the question is interpreted as allowing the trivial ideal $\{0\}$, the same conclusion holds: the canonical map has all of E as kernel. Under the paper's standing non-trivial-ideal convention, the one-dimensional example $E = \mathbb{R}$ admits no candidate proper ideal at all.

6 Novelty check and limitations

On 9 August 2026, a bounded search covered the run indexes, arXiv:2207.05459, the exact question, and combinations of “proper ideal”, “projection bands”, “Dedekind complete vector lattice”, and “inverse limit”. It found the source and published versions of the paper, but no later paper explicitly answering the question.

The result settles only the literal universal statement. It does not answer the natural repaired question in which E is required to be infinite-dimensional. The source examples surrounding the question are infinite-dimensional, so this omitted hypothesis may have been intended. That scope issue lowers the mathematical novelty of the observation but does not affect the counterexample to the printed statement.

Human-review recommendation. Check the intended quantifier and scope. If the question is read as printed, the kernel argument is complete.

References

[1] W. van Amstel and J. H. van der Walt, *Limits of vector lattices*, arXiv:2207.05459 (2022; revised 2023), Remark 4.12 and page 23, https://arxiv.org/abs/2207.05459.